

# Corollaries of the Watermark Theorem

*The transcendental lattice and the Brauer group of Fermat surfaces of prime degree*

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## Abstract

The Watermark Theorem [T] gives, for every prime  $m \geq 5$ , the exact discriminant of the Néron-Severi lattice of the complex Fermat surface  $S_m$ :  $|\text{disc NS}(S_m)| = m^{3(m-3)^2}$ . We record here what falls out immediately. Corollary 1, fully rigorous and five lines long, computes the discriminant of the transcendental lattice:  $|\text{disc } T_m| = m^{3(m-3)^2}$ , on rank  $(m-1)(m-3)^2$ . Corollary 2 pins the order of the discriminant group  $\text{NS}^*/\text{NS}$ . Corollary 3 assembles the closed-form recipe for the order of the Brauer group of the good reductions of  $S_m$  over finite fields via the Artin-Tate formula — a formula whose every ingredient (the Frobenius eigenvalues, classically the Jacobi sums; the Néron-Severi discriminant, now the Theorem; the elementary  $q$ -powers) is explicit. The status of each statement — proven here, classical input, or evaluation left per  $(q, m)$  — is labelled line by line.

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## 1. Setup

$S_m \subset \mathbb{P}^3$  is the Fermat surface of prime degree  $m \geq 5$ .  $H^2(S_m, \mathbb{Z})$  is unimodular (Poincaré duality) and torsion-free ( $S_m$  is a smooth hypersurface in  $\mathbb{P}^3$ , hence simply connected), of rank  $b_2 = m^3 - 4m^2 + 6m - 2$ .  $\text{NS} = \text{NS}(S_m)$  is a primitive sublattice (it is the intersection of  $H^2$  with  $H^{\{1,1\}}$ , hence saturated), of rank  $\rho = 3(m-1)(m-2) + 1$  [Aoki-Shioda; Schütt-Shioda-van Luijk], and the Watermark Theorem gives  $|\text{disc NS}| = m^{3(m-3)^2}$ . The transcendental lattice is  $T_m := \text{NS}^\perp$  in  $H^2(S_m, \mathbb{Z})$ .

## 2. Corollary 1 — the transcendental lattice

$$|\text{disc } T_m| = m^{3(m-3)^2}, \text{ and rank } T_m = (m-1)(m-3)^2.$$

**Proof.** For a primitive sublattice  $N$  of a unimodular lattice  $U$  with orthogonal complement  $T$ , the discriminant groups  $N/N$  and  $T/T$  are canonically isomorphic (anti-isometrically, by standard lattice theory: gluing inside  $U$ ). Hence  $|\text{disc } T_m| = |\text{disc NS}| = m^{3(m-3)^2}$  by the Theorem. The rank:  $b_2 - \rho = (m^3 - 4m^2 + 6m - 2) - (3m^2 - 9m + 7) = m^3 - 7m^2 + 15m - 9 = (m-1)(m-3)^2$ . ■

*Checks:*  $m = 5$ :  $b_2 = 53$ ,  $\rho = 37$ ,  $\text{rank } T = 16 = 4 \cdot 2^2$ ,  $\text{disc } 5^{12}$ .  $m = 7$ :  $b_2 = 187$ ,  $\rho = 91$ ,  $\text{rank } T = 96 = 6 \cdot 4^2$ ,  $\text{disc } 7^{48}$ .  $m = 11$ :  $\text{rank } T = 640$ ,  $\text{disc } 11^{192}$ .

**Remark.** The exponent  $3(m-3)^2$  is three times the square of  $m-3$  — and  $(m-3)^2$ , in Shioda's framework, counts a basic family of transcendental character orbits. Corollary 1 places that square on the transcendental side of the unimodular wall with a proof, where [7] could place it only conjecturally.

## 3. Corollary 2 — the discriminant group

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$$|\mathrm{NS}/\mathrm{NS}| = |\mathrm{T}/\mathrm{T}| = m^{3(m-3)^2}; \text{ both groups are } m\text{-groups.}$$

**Proof.** The order is the discriminant; the group is killed by a power of  $m$  since the discriminant is. ■

The finer structure (the elementary divisors) is *measured, not proven*: the campaign's instrument rail shows the nonzero elementary-divisor profile of the line Gram coincides with the rows tabulated by Aljovin-Movasati-Villaflor for low degree (e.g.  $5^{10} \cdot 25$  at  $m = 5$ ;  $7^{38} \cdot 49^5$  at  $m = 7$ ), and the campaign's profile law ( $\#(m^2\text{-divisors}) = 3m - 16$  for prime  $m \geq 7$ ) fits all published rows — catalogued in [W], conjectural beyond the measured range.

#### 4. Corollary 3 — the Brauer group over finite fields: the closed recipe

Fix a prime power  $q = p^r$  with  $p \nmid m$ , and let  $X = S_m \otimes F_q$  be the (smooth) reduction. The following inputs are classical:

- **(Tate for Fermat surfaces.)** The Tate conjecture holds for Fermat surfaces over finite fields (Tate's original cases and the inductive structure of Shioda-Katsura, *On Fermat varieties*, Tôhoku Math. J. 31 (1979); see also Tate, *Algebraic cycles and poles of zeta-functions*, 1965). Consequently  $\mathrm{Br}(X)$  is finite and the Artin-Tate formula holds (Tate, Bourbaki 1966; Milne, Ann. of Math. 102 (1975)).
- **(Frobenius eigenvalues = Jacobi sums.)** The eigenvalues of Frobenius on  $H^2_{\text{ét}}(\tilde{X}, \mathbb{Q}_\ell)$  are  $q$  times roots of unity on the algebraic part and the Jacobi-sum characters  $j(\chi)$  on the transcendental part (Weil, 1949) — explicitly computable for each  $(q, m)$ .
- **(Torsion.)**  $\mathrm{NS}(X)$  is torsion-free for Fermat surfaces.

**The Artin-Tate formula** (in Milne's normalization) then reads, for  $X$  with  $b_1 = 0$ :

$$\lim_{s \rightarrow 1} P_2(X, q^{-s}) / (1 - q^{1-s})^{\{\rho(X)\}} = \pm |\mathrm{Br}(X)| \cdot |\mathrm{disc} \mathrm{NS}(X)| / (q^{\{\alpha(X)\}} \cdot |\mathrm{NS}(X)_{\text{tors}}|^2), \quad \alpha(X) = \chi(O_X) - 1,$$

where  $P_2$  is the degree- $b_2$  Frobenius factor of the zeta function and  $\rho$  the Néron-Severi rank over  $F_q$ . The Brauer order is thus the special value of  $P_2$  at  $s = 1$  divided by the Néron-Severi discriminant, up to the explicit  $q$ -power; for Fermat surfaces  $\mathrm{NS}$  is torsion-free. The precise statement used is the Artin-Tate formula as established in Milne (1975); we do not restate its proof.

**Corollary 3.** *For each  $q \equiv 1 \pmod{m}$  such that  $\mathrm{NS}(\tilde{X})$  is defined over  $F_q$  (full descent of the lines), the order  $|\mathrm{Br}(X)|$  is given in closed form by the Artin-Tate special value with  $|\mathrm{disc} \mathrm{NS}(X)| = m^{3(m-3)^2}$  supplied by the Watermark Theorem, the transcendental eigenvalues supplied by Weil's Jacobi sums, and  $\chi(O_X) = 1 + p_g = 1 + (m-1)(m-2)(m-3)/6$ . Every ingredient is explicit; the evaluation is finite arithmetic per  $(q, m)$ .*

**Status, line by line.** *Proven here:* the discriminant input (the Theorem) and the rank/holomorphic-genus arithmetic. *Classical, cited:* Tate for Fermat surfaces; the Artin-Tate formula; Weil's eigenvalue computation; the squareness of  $|\mathrm{Br}|$  (Milne for  $q$  odd, completed in later literature). *Left explicit, per case:* the descent condition ( $q \equiv 1 \pmod{m}$  with the lines rational over  $F_q$  — under which  $\mathrm{NS}(X) = \mathrm{NS}(\tilde{X})$  and  $\mathrm{disc} \mathrm{NS}(X)$  is the complex value), and the numerical evaluation of the Jacobi-sum product. No Brauer order is asserted for any specific  $(q, m)$  in this

note; what is asserted is that the recipe is now closed — before the Theorem, its discriminant denominator was conjectural for every prime  $m$  beyond the verified range.

## 5. What is *not* claimed

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The structure of  $\text{Br}(X)$  (beyond its order through the recipe), the supersingular cases  $p$  with special congruences, odd composite  $m$ , and the even half of the Watermark Law are outside this note. The complex (cohomological) Brauer group of  $S_m$  has divisible part of corank rank  $T_m = (m-1)(m-3)^2$ ; its torsion-theoretic fine structure is not addressed here.

## References

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[T] R. Amichis Luengo, *The Watermark Theorem* (this repository: THE\_WATERMARK\_THEOREM.md / .pdf). [W] *The Watermark Law* (this repository: THE\_WATERMARK\_LAW.md). 1. T. Shioda, *Some observations on Jacobi sums*. Adv. Stud. Pure Math. **12** (1987), 119–135. 2. M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*. J. Number Theory **130**:9 (2010), 1939–1963. 3. N. Aoki, T. Shioda, *Generators of the Néron-Severi group of a Fermat surface*. Progr. Math. **35** (1983), 1–12. 4. T. Shioda, T. Katsura, *On Fermat varieties*. Tôhoku Math. J. **31** (1979), 97–115. 5. J. Tate, *Algebraic cycles and poles of zeta-functions*. In: Arithmetical Algebraic Geometry, Harper & Row (1965). 6. J. Tate, *On the conjectures of Birch and Swinnerton-Dyer and a geometric analog*. Séminaire Bourbaki **306** (1966). 7. J. S. Milne, *On a conjecture of Artin and Tate*. Ann. of Math. **102** (1975), 517–533. 8. A. Weil, *Numbers of solutions of equations in finite fields*. Bull. Amer. Math. Soc. **55** (1949), 497–508.

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*Corollary 1 is a complete proof. Corollary 3 is a closed recipe with classical inputs cited and every conditional stated. Nothing here asks for trust — it asks to be checked.*